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SMART FAN - Detecting Human Presence

Faculty Mentor:
Dr. Annamraju Sai Srikar

PROBLEM STATEMENT

Traditional table fans provide fixed airflow and require manual adjustment, resulting in inefficient cooling and energy wastage; hence, our team is set to develop a smart fan that automatically tracks user movement and adjusts airflow direction and speed based on real-time temperature.

OBJECTIVES

1. Human Motion Detection
To implement sensors that can detect and track user movement within a defined range.

2. Targeted Airflow Control
To develop a mechanism that allows the fan to rotate dynamically to follow the detected user position in real time.

3. Temperature Monitoring System
To integrate temperature sensors that continuously measure room conditions.

4. Automatic Speed Regulation
To create an algorithm that adjusts fan speed based on temperature changes for optimal comfort.

5. User-Friendly Interface
To provide simple controls (buttons/app/voice) for manual override and customization.

CONCEPT DESIGN

1. Detection & Tracking: A PIR sensor monitors motion. In Standby, the fan oscillates (45°–135°) to scan the area. Upon Detection, the servo locks onto the person's position instantly.
2. Adaptive Cooling: The system receives temperature data via a DAC/ADC bridge. It automatically adjusts PWM Duty Cycle to scale fan speed based on ambient heat (Faster = Hotter).
3. Intelligent Timing: A 20-second hold timer prevents frequent toggling, ensuring the fan stays focused even during brief periods of inactivity.
4. IoT Monitoring: Hosts a local Wi-Fi Access Point with a live telemetry dashboard for real-time speed and temperature tracking.

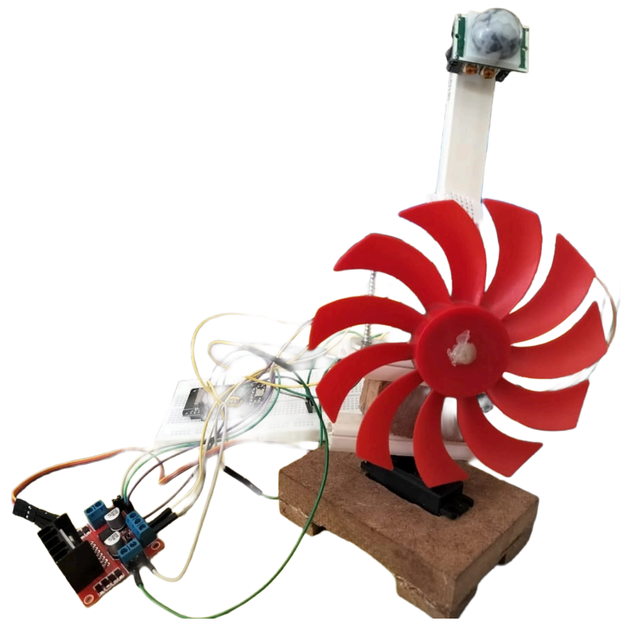
INDIVIDUAL CONTRIBUTIONS

- Adithya Ajay - Sensor Integration
- Hemanthraj M - Coding and Poster making
- S Abhishek - Interfacing and Embedded Coding
- Antony - Mechanical Design and Assembly
- Gautam - Motor and Actuation Control checking
- Vishnu - Hardware and Mechanical design
- Pranav - Testing and Validation

REFERENCES

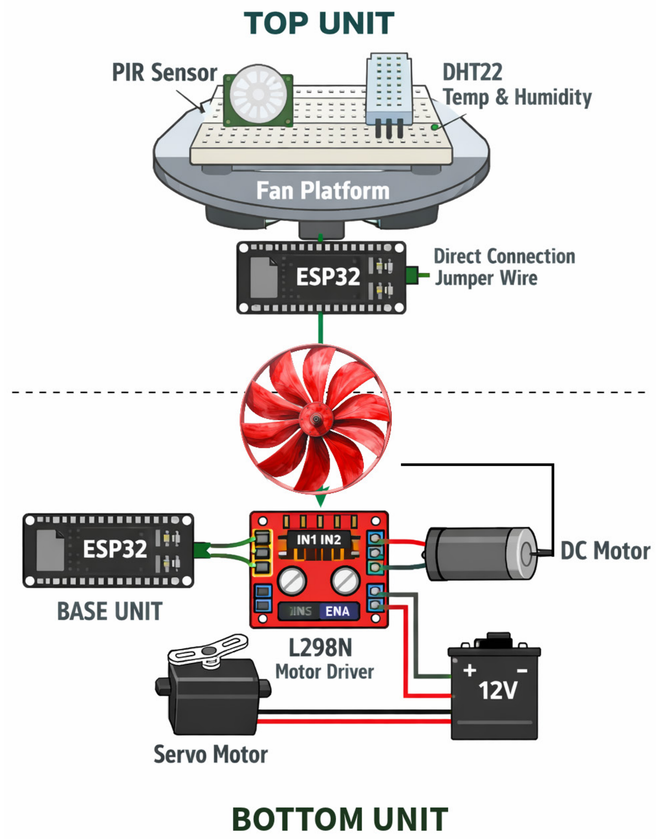
T. Ishrat, M. A. Rahaman and A. Ahammad, "Smart fan for human tracking," 2014 9th International Forum on Strategic Technology (IFOST), Cox's Bazar, Bangladesh, 2014, pp. 124-127, doi: 10.1109/IFOST.2014.6991086. keywords: {Acoustics;Temperature sensors;Robot sensing systems;Temperature dependence;Target tracking;Temperature measurement;human tracking;ultrasonic distance sensor;temperature sensor;micro controller;lcd display},

PROTOTYPE



Limitations:
1. Delay: ~200–600 ms (normal)
2. PIR: detects motion, NOT people
3. Servo: no real angle control
4. Analog signal = noisy
5. Power system must be stable

Conceptual Diagram



OUTCOMES AND INFERENCES

1. Smart automation significantly enhances cooling efficiency and convenience.
2. Targeted airflow helps avoid unnecessary energy usage.
3. Sensor-based systems can effectively replace manual adjustments in everyday appliances, especially for the elderly.
4. Real-time environmental adaptation leads to better comfort optimization.
5. The concept demonstrates strong potential for future smart home applications.

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